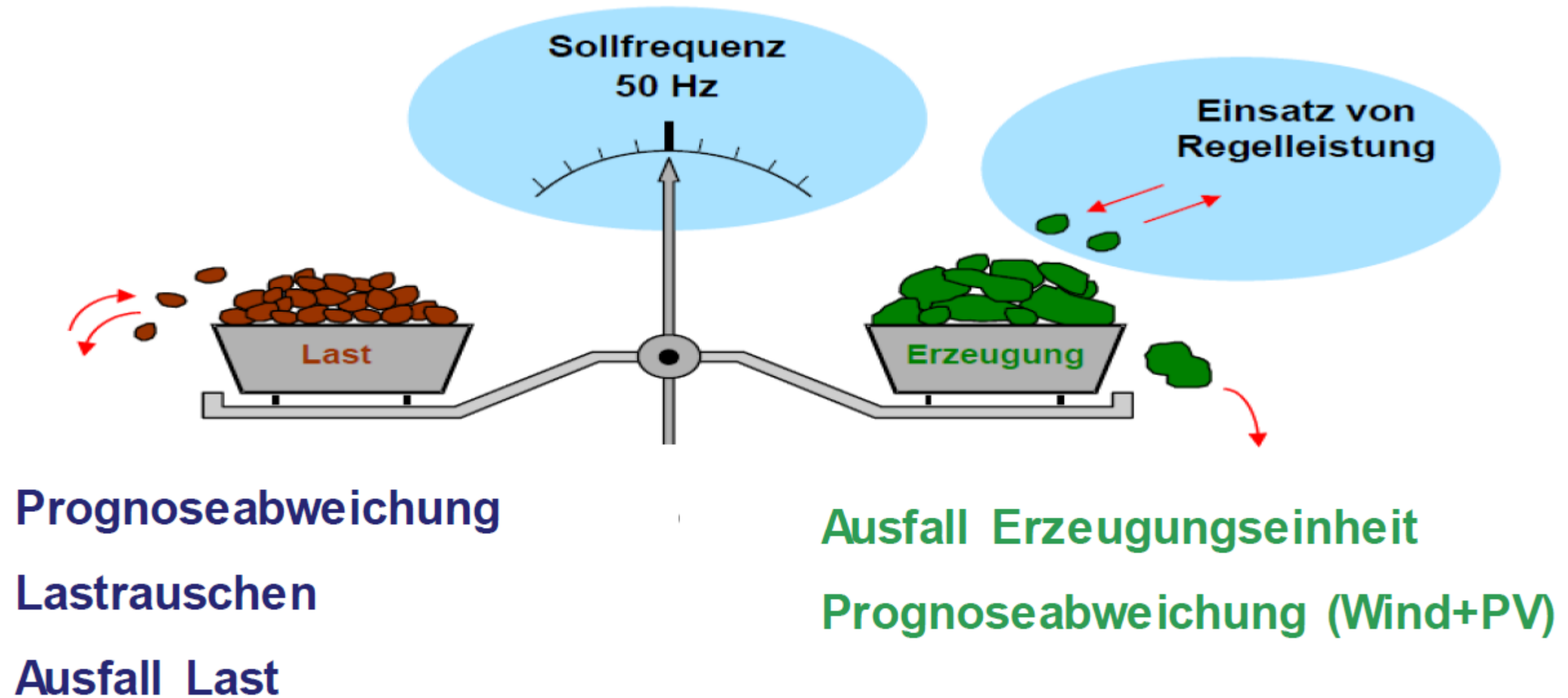


Power System Control

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Course Number 47083

Introduction

- ♦ Production and Consumption needs to be balanced



Source: Hagen/Schöller

Introduction

- ◆ ENTSO-E
 - Control area manager (CAM) (dt. Regelzonenführer)
 - Monitors their area
 - Provide appropriate countermeasure

- ◆ Reasons for un-balance
 - Power plant failure
 - Grid failure (Storm, accidents)
 - Inaccuracy in the load forecast
 - Inaccuracy in the production forecast
 - Unexpected metrological events

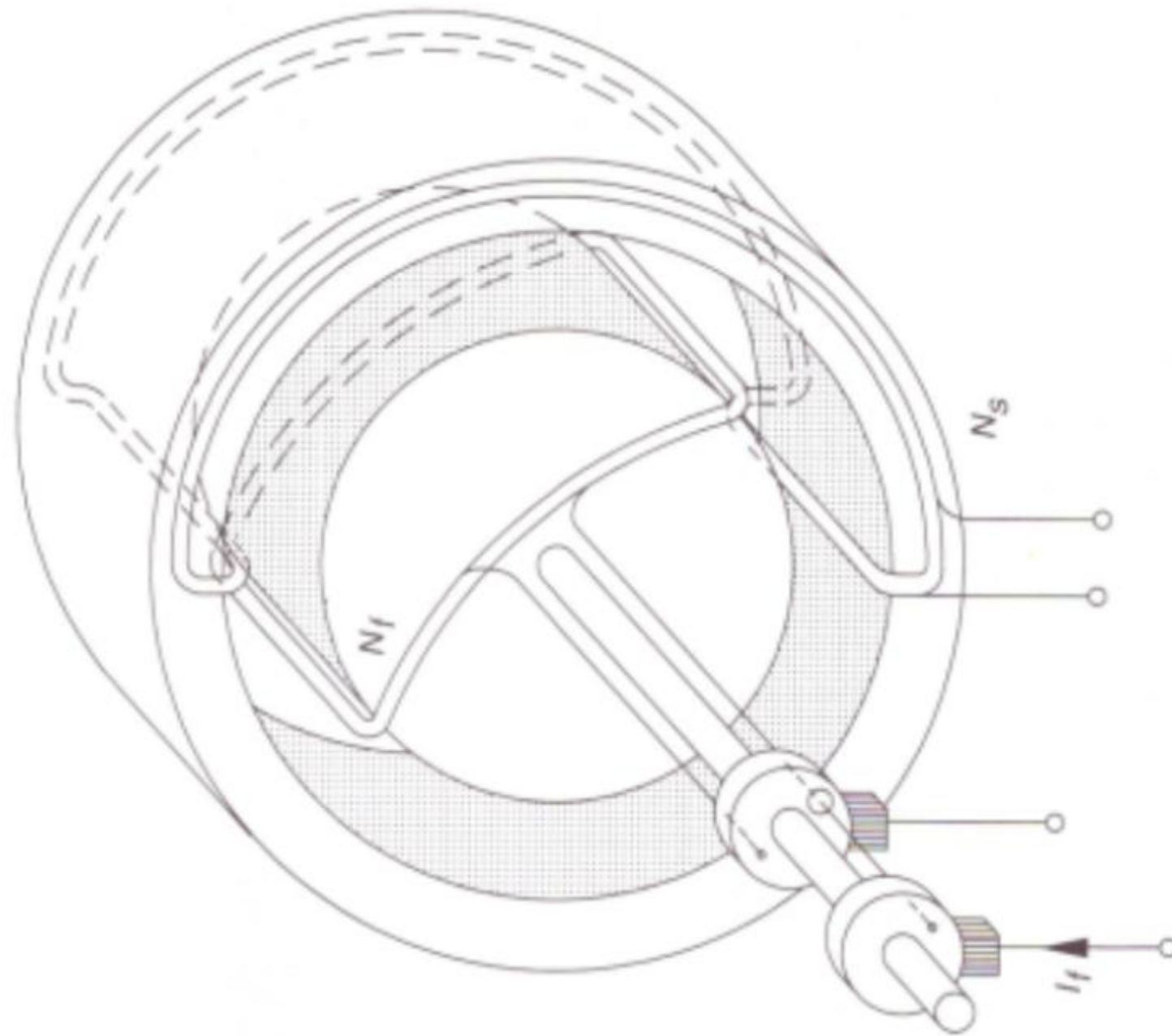
Introduction

- ◆ TSO's have the role of CAM
 - Purchase system services
 - Real power
 - Reactive power
 - Open, market based procedure
 - Transparent
 - Non discriminatory

Synchronous Generators

- ◆ Majority of electrical energy is produced by three-phase synchronous generators
- ◆ Rotor
 - DC-excited winding
 - Acts as electromagnet
- ◆ Function
 - Rotor rotates
 - rotating magnetic field generates a time-varying electromotive force EMF (Volts) in the three-phase stator windings
 - Stator windings are directly connected with the grid
- ◆ All synchronous generators in the ENTSO-Grid are connected
 - Rotating mass

Synchronous Generators



Frequency Support

- ♦ Grid **frequency** directly related to **speed** of the synchronous generators
- ♦ More electrical load applied
 - Generator reduces its speed
- ♦ As load is removed
 - Generator will speed up
- ♦ Grid frequency is increased by adding generation power

Balancing within Continental European Grid

- ◆ Three stage procedure
 - Primary Control
 - Secondary Control
 - Tertiary Control
- ◆ CAM (TSO) are in charge if this procedure

Primary Control

- ♦ Restore the balance within few seconds
- ♦ Activation done directly by the turbine's speed governor
 - e.g. throttle valve at a thermal power plant
- ♦ Time period for a single incident
 - $0 < t < 30\text{s}$
- ♦ All TSOs in the ENTSO-E are participating
 - Primary power capacity +/- 3000 MW
 - Volume provided by APG appr. +/- 65 MW

Secondary control

- ♦ Primary control does not get the frequency back to the nominal value
- ♦ Automatically actuated by the central grid controller within the **control area** (e.g. Austria) where the power got lost.
- ♦ Relieves the primary control outside this area
- ♦ Controller has proportional-integral characteristic – brings frequency back to nominal value
- ♦ Power plants participating in secondary control
 - Are in operation at around 75% of their nominal capacity
 - Able to increase / decrease capacity
- ♦ Time period for a single incident
 - $30\text{ s} < t < 15\text{ min}$

Tertiary Control

- ♦ Tertiary control is called minutes reserve
- ♦ Used to relief secondary control
- ♦ In APG control area
 - Activated manually from dedicated suppliers
- ♦ Time period for a single incident
 - $t < 15$ min up to several hours

Voltage Support

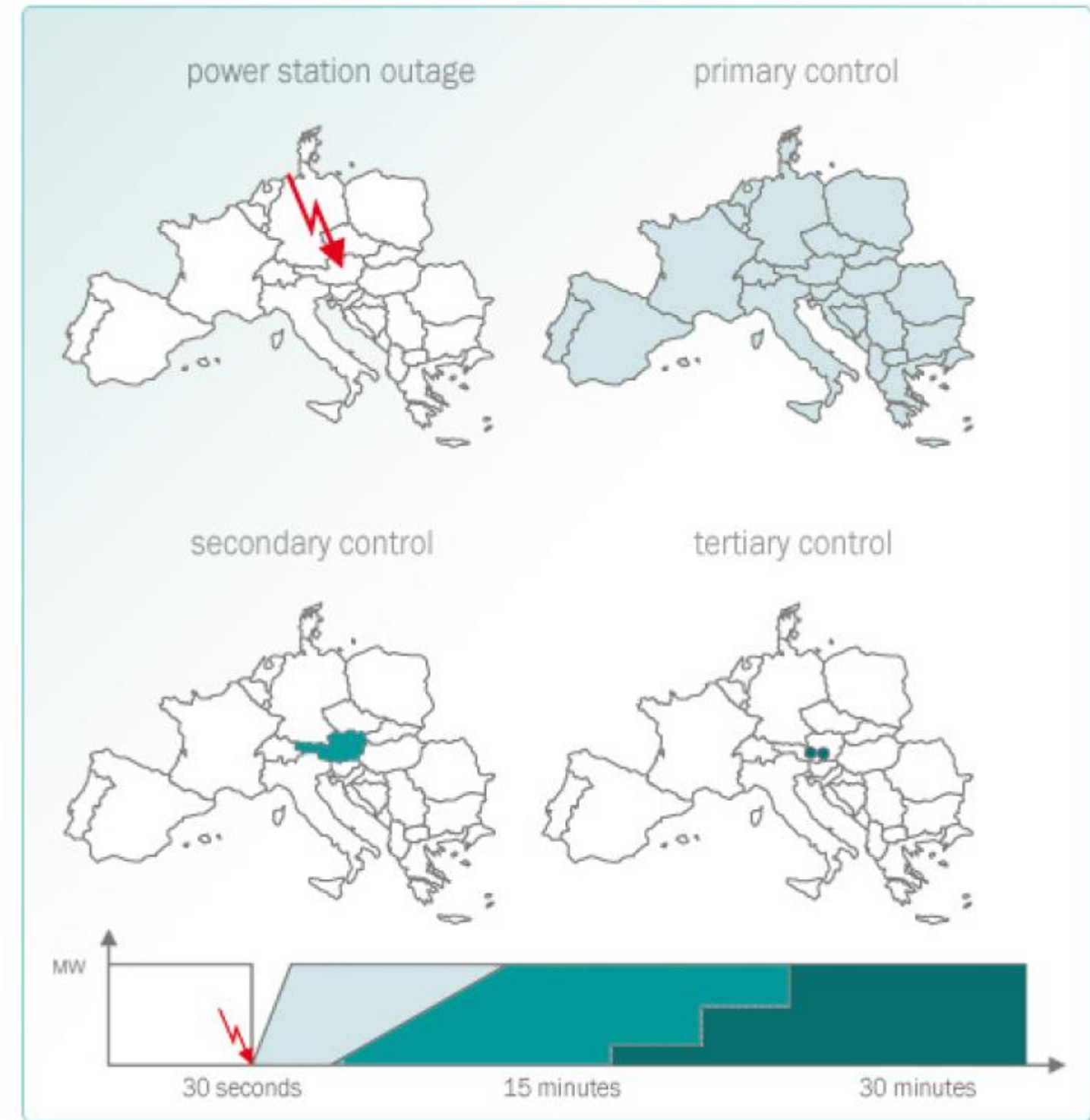
- ◆ Power grid Frequency relates to the balance of real power
- ◆ Power grid Voltage depends on the reactive power balance
- ◆ Reactive power consumption increases
 - Voltage drops
- ◆ Reactive power consumption decreases
 - Voltage increases
- ◆ Power station must provide mandatory reactive power at any time
- ◆ Additionally the CAM has contractual agreements for the supply of reactive capacities

Example from Book: Giesecke and Heimerl 2014

- ◆ Fault happens in a power plant (nominal output power 100 MW)
- ◆ Plant gets disconnected
 - Grid is out of balance immediately
 - Frequency change of 0.01 Hz
- ◆ Primary control immediately starts adding power
- ◆ Within few seconds balance is established
- ◆ Secondary controller starts to bring frequency back to nominal
- ◆ After appr. 10 to 30 seconds the pump storage power plants start producing power
- ◆ Pump storage plant reach its rated output within 1 to 3 minutes
- ◆ Reserve power plant is started up to relive pump storage power plant

Summary

- ♦ Major in-balance because of power station outage
- ♦ Primary control immediately reacts
- ♦ Disturbance not cleared ($< 30s$)
 - Secondary control (APG)
- ♦ Disturbance not cleared ($< 15 \text{ min}$)
 - Tertiary control (APG)
 - Power plants activated





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